NORTHERN WOLVES
AIR CONDITIONING

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Customer
Procida Construction Corp.
Attn: Richard Jinks

Project: **India Home Senior Housing - Site 1**

Address: **170-11 89th Avenue, Jamaica, Queens, NY**

Mechanical drawings dated: 07.31.2026 (pricing set, M-100.00 to M-109.00)

Mechanical Value Engineering Proposals

At the owner's request, Northern Wolves reviewed the mechanical pricing set and proposes the value engineering items below. The apartments are designed with through-wall PTHPs and central roof ERVs with shaft risers and two-position airflow regulators in every apartment; the common areas use VRF and split systems with separate ERVs. The largest savings are in the apartment ventilation, which repeats on every floor (VE-01 and VE-02). Each item can be accepted on its own. Deduct pricing will follow once Procida and the owner select the items to pursue.

VE	Item	Area	Approvals
VE-01	Apartment exhaust: one exhaust riser per shaft (kitchen and bath combined)	Floors 2 to 9, shaft stacks 1 to 5; roof ERV-1 and ERV-2	Altera Engineering (riser sizing, ERV exhaust); RKTB (shaft sizes).
VE-02	Apartment supply: constant-airflow regulators in lieu of two-position regulators	All apartments, floors 2 to 9 (53 ZRT-2 supply regulators)	Altera Engineering; energy / Passive House consultant if the project is certified (ventilation balance during boost).
VE-03	1st floor VRF: heat pump system in lieu of heat recovery	1st floor common areas (ACCU-1.1, ACCU-1.2, HRU-1, AC-1.1 to AC-1.5)	Altera Engineering (zoning and load check); owner.
VE-04	Linear slot diffusers: standard ceiling diffusers	1st and 2nd floor common areas (LDS / LDR)	RKTB (ceiling design); Altera Engineering (throw at the glazing).
VE-05	PTHP condensate drains: delete heat tracing; plastic drain piping where permitted	Apartment PTHP-1 / PTHP-2, floors 2 to 9; 2nd floor exposed drain drops	Altera Engineering; RKTB (chase locations).
VE-06	Laundry: ventless heat-pump dryers (owner option)	Laundry room; (4) 4" dryer exhaust ducts to the roof	Owner (laundry equipment); RKTB; condensate drain by plumbing.
VE-07	Cellar service rooms: ventilation in lieu of split AC where the load allows	Cellar - AC-COMP and AC-EMR rooms (AC-TEL kept)	Altera Engineering (room use and heat loads); EE.
VE-08	Cellar and 1st floor ventilation: one ERV in lieu of two	ERV-C (cellar) and ERV-CF (1st floor)	Altera Engineering (airflow, fire separation); owner, if the cellar and 1st floor are separate tenants.

Notes

- Deduct amounts will be provided for the VE items selected by Procida and the owner.
- Cut sheets will be provided with the pricing for the items that change equipment.
- VE items require the approval of the engineer of record; redesign, revised drawings and any DOB amendment by others.

- Site 2, 150-05 88th Avenue (Quote 3839), uses the same systems; the same items are expected to apply there. A separate review will follow on receipt of the Site 2 drawings.

VE-01 Apartment exhaust: one exhaust riser per shaft (kitchen and bath combined)

Area: Floors 2 to 9, shaft stacks 1 to 5; roof ERV-1 and ERV-2

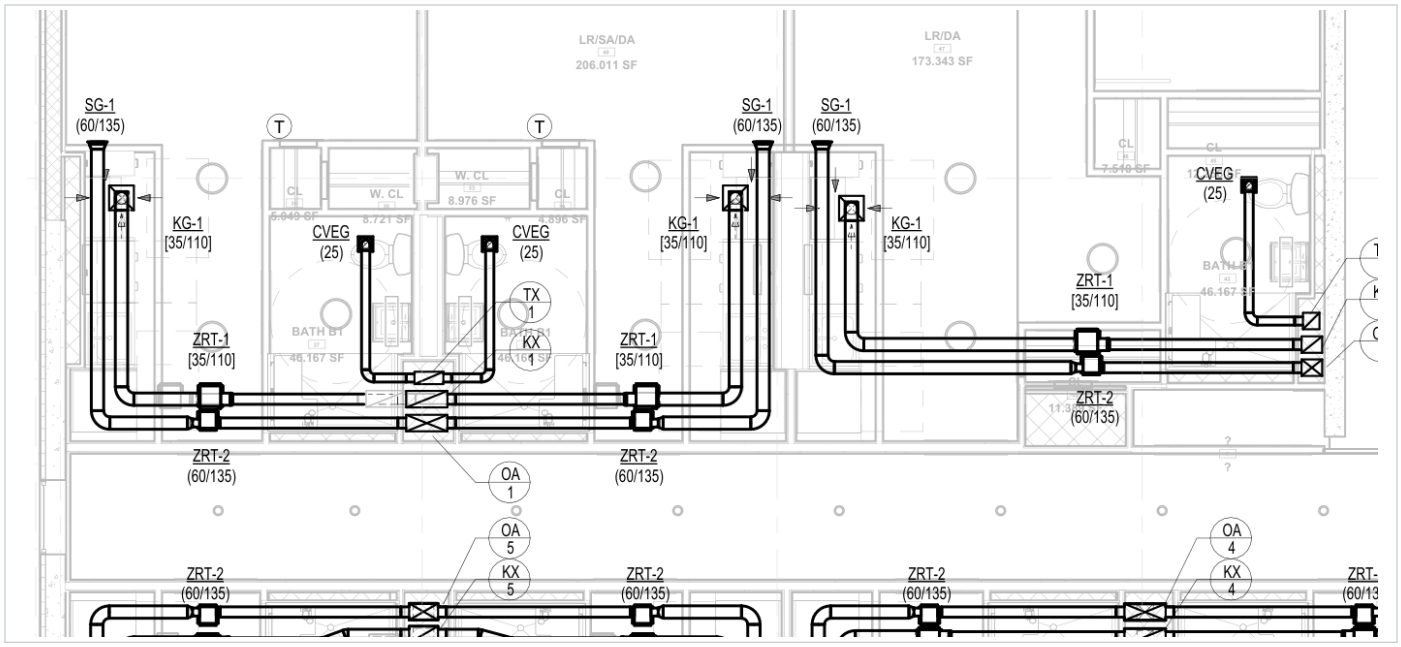
As designed

- Each of the 5 shaft stacks carries 3 risers: OA (supply), KX (kitchen exhaust) and TX (toilet exhaust), 15 risers from the 2nd floor to the roof ERV-1 / ERV-2
- Kitchen exhaust KG-1 on ZRT-1 (35/110 CFM) to the KX riser; bath exhaust CVEG (25 CFM) to the TX riser

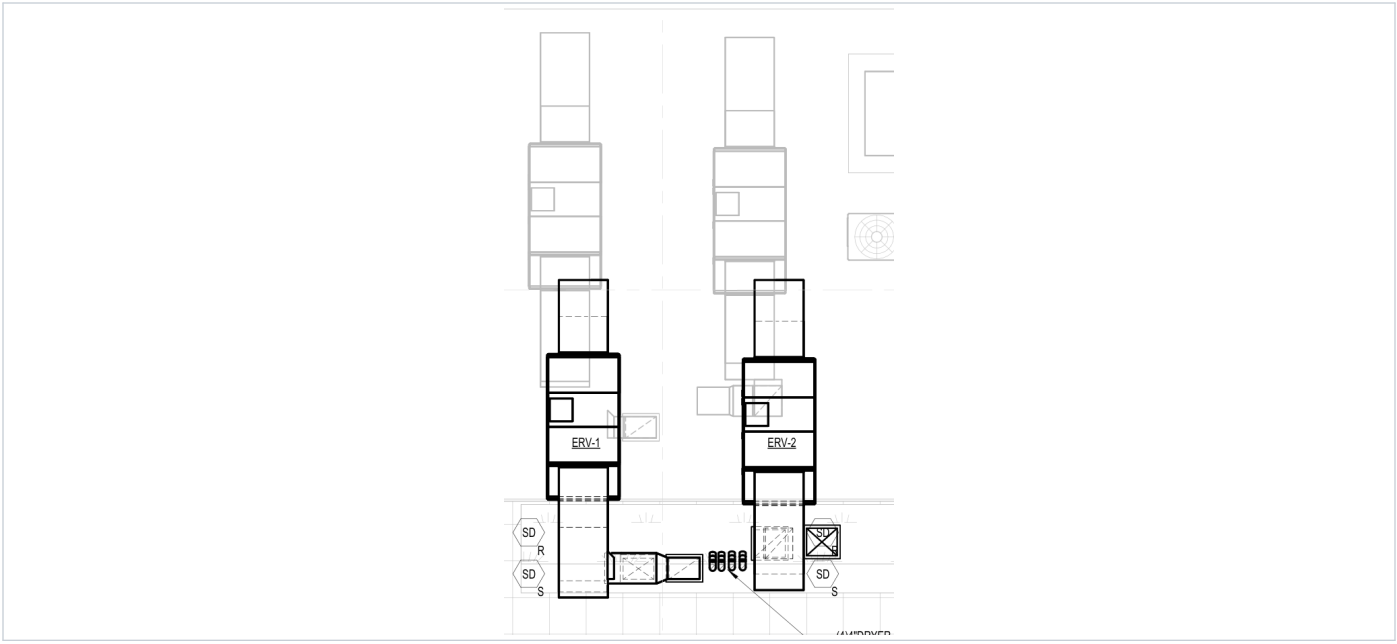
Proposed

- Kitchen and bath exhaust branches of each apartment connect to one exhaust riser per stack, sized for the combined airflow: 5 exhaust risers in lieu of 10
- Same ERVs, grilles and regulators; KG-1 kitchen grilles with washable grease filters
- One exhaust connection per stack at the roof

Benefits: 5 fewer full-height risers (2nd floor to roof), fewer roof penetrations and ERV connections, smaller shafts (floor area may be returned to the apartments).
Electrical impact: None
Approvals / coordination: Altera Engineering (riser sizing, ERV exhaust); RKTB (shaft sizes).



M-103.00 Typical 3rd to 5th floor - OA / KX / TX risers at stack 1, ZRT-1 / ZRT-2 regulators



M-108.00 Roof - ERV-1 and ERV-2

VE-02 Apartment supply: constant-airflow regulators in lieu of two-position regulators

Area: All apartments, floors 2 to 9 (53 ZRT-2 supply regulators)

As designed

- Each apartment has two two-position regulators: ZRT-1 on kitchen exhaust (35/110 CFM) and ZRT-2 on supply SG-1 (60/135 CFM), with boost control and wiring; 106 regulators in total

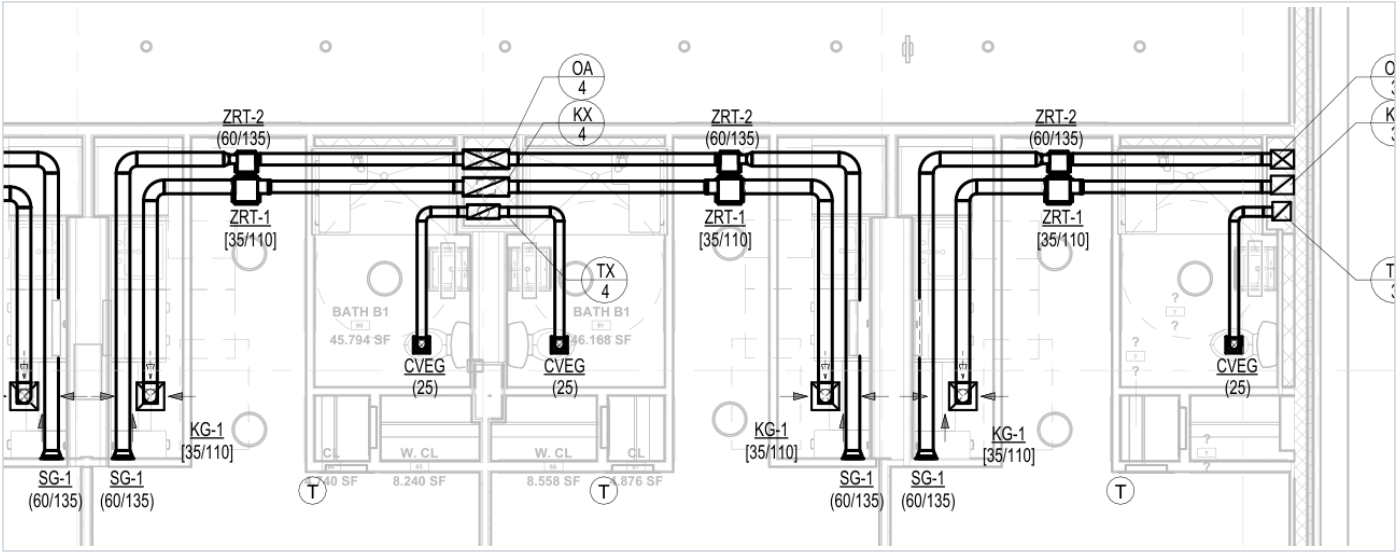
Proposed

- Replace the 53 ZRT-2 supply regulators with passive constant-airflow regulators set to 60 CFM: no motor, no wiring, no controls
- Keep ZRT-1 with kitchen boost. During boost the apartment runs slightly exhaust-heavy for the boost period

Benefits: 53 fewer motorized regulators with their control wiring and boost interlocks; simpler balancing; nothing to maintain on the supply side.

Electrical impact: Deletes 53 regulator power / control connections

Approvals / coordination: Altera Engineering; energy / Passive House consultant if the project is certified (ventilation balance during boost).



M-103.00 Typical floor - SG-1 supply on ZRT-2 (60/135) and KG-1 kitchen exhaust on ZRT-1 (35/110) in each apartment

VE-03 1st floor VRF: heat pump system in lieu of heat recovery

Area: 1st floor common areas (ACCU-1.1, ACCU-1.2, HRU-1, AC-1.1 to AC-1.5)

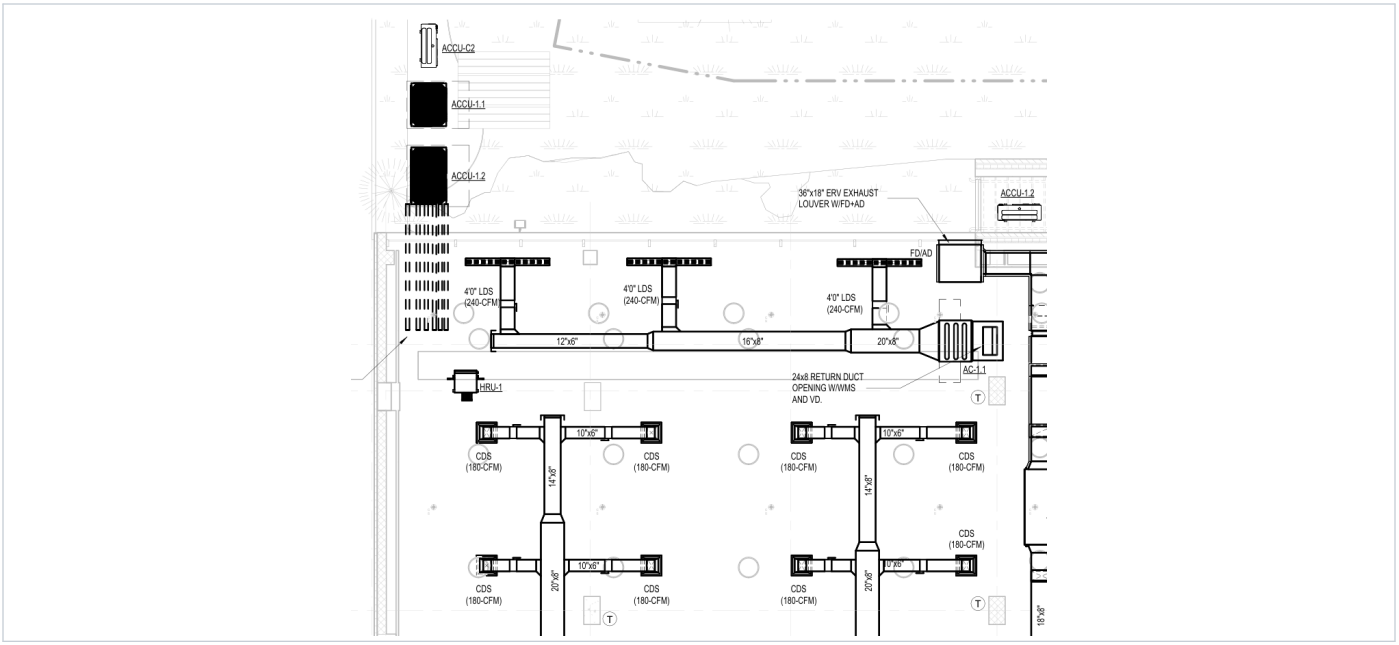
As designed

- VRF heat recovery system: outdoor units ACCU-1.1 / ACCU-1.2 with heat recovery unit HRU-1 and 3-pipe refrigerant piping to the ducted indoor units AC-1.1 to AC-1.5

Proposed

- VRF heat pump system (2-pipe), no heat recovery unit; same indoor units, ductwork and air outlets
- The areas served are open, connected common spaces that do not need heating and cooling at the same time

Benefits: Deletes HRU-1, the third refrigerant pipe with its insulation and the heat recovery controls; simpler service.
Electrical impact: None
Approvals / coordination: Altera Engineering (zoning and load check); owner.



M-101.00 1st floor - ACCU-1.1 / ACCU-1.2, HRU-1 and ducted indoor units

VE-04 Linear slot diffusers: standard ceiling diffusers

Area: 1st and 2nd floor common areas (LDS / LDR)

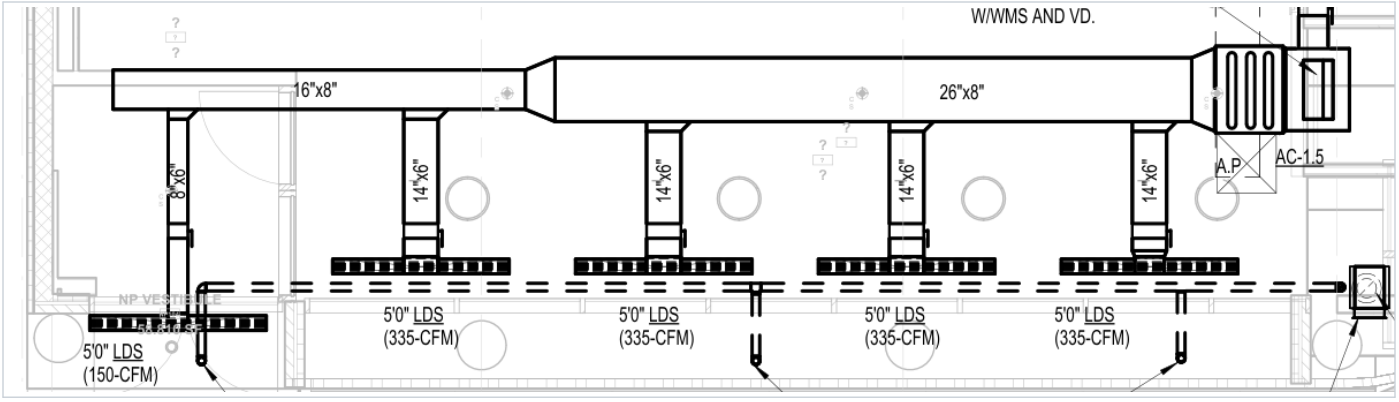
As designed

- Linear slot supply diffusers 4'-0" and 5'-0" LDS (about 16), 6'-0" LDR returns (5) and a 12'-0" return linear diffuser, with plenums

Proposed

- Standard louvered ceiling diffusers and return grilles at the same airflows; sidewall grilles where the architect prefers

Benefits: Lower-cost air outlets, no plenums, faster installation and balancing.
Electrical impact: None
Approvals / coordination: RKTB (ceiling design); Altera Engineering (throw at the glazing).



M-101.00 1st floor - 5'-0" LDS along the storefront on AC-1.5

VE-05 PTHP condensate drains: delete heat tracing; plastic drain piping where permitted

Area: Apartment PTHP-1 / PTHP-2, floors 2 to 9; 2nd floor exposed drain drops

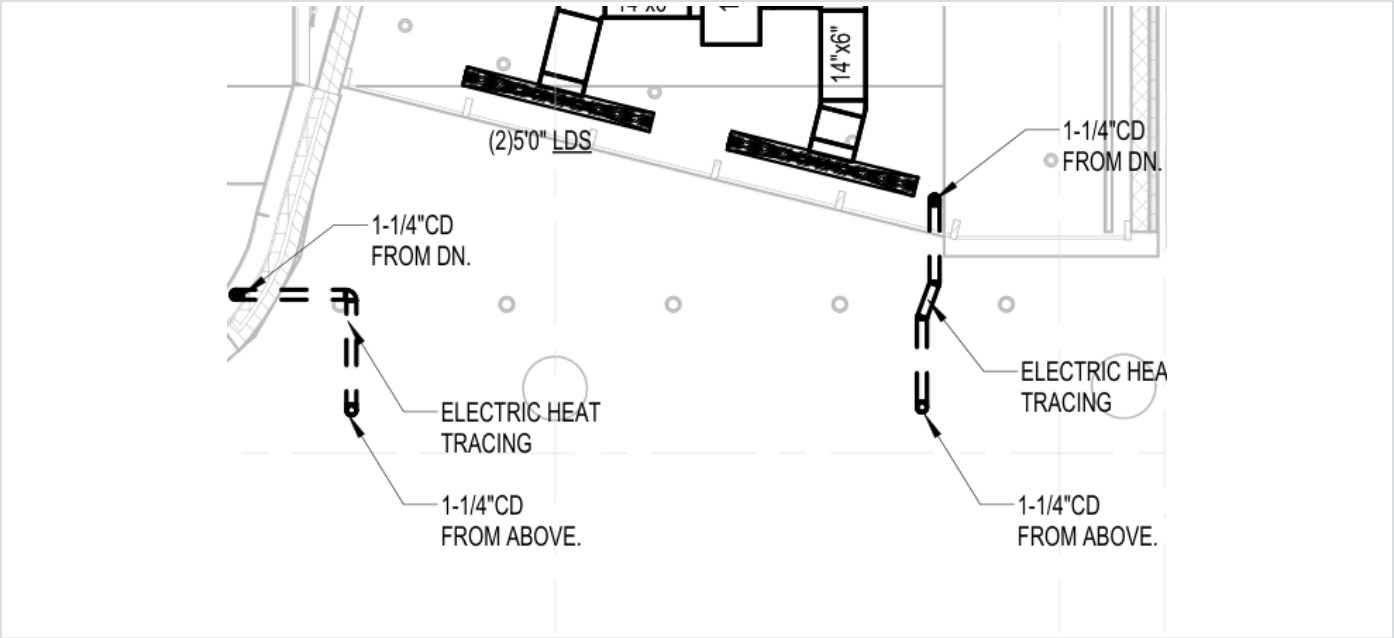
As designed

- 1-1/4" condensate drain at each PTHP; at the 2nd floor the drain drops run in an exposed location with electric heat tracing

Proposed

- Route the 2nd floor drain drops inside the heated building envelope: no heat tracing
- PVC / CPVC condensate piping where permitted by the NYC Mechanical Code and the specification, if copper is specified

Benefits: Deletes the heat tracing with its circuits and controls (electrical scope); lower material cost.
Electrical impact: Deletes the heat tracing circuits
Approvals / coordination: Altera Engineering; RKTB (chase locations).



M-102.00 2nd floor - 1-1/4" CD drops with electric heat tracing

VE-06 Laundry: ventless heat-pump dryers (owner option)

Area: Laundry room; (4) 4" dryer exhaust ducts to the roof

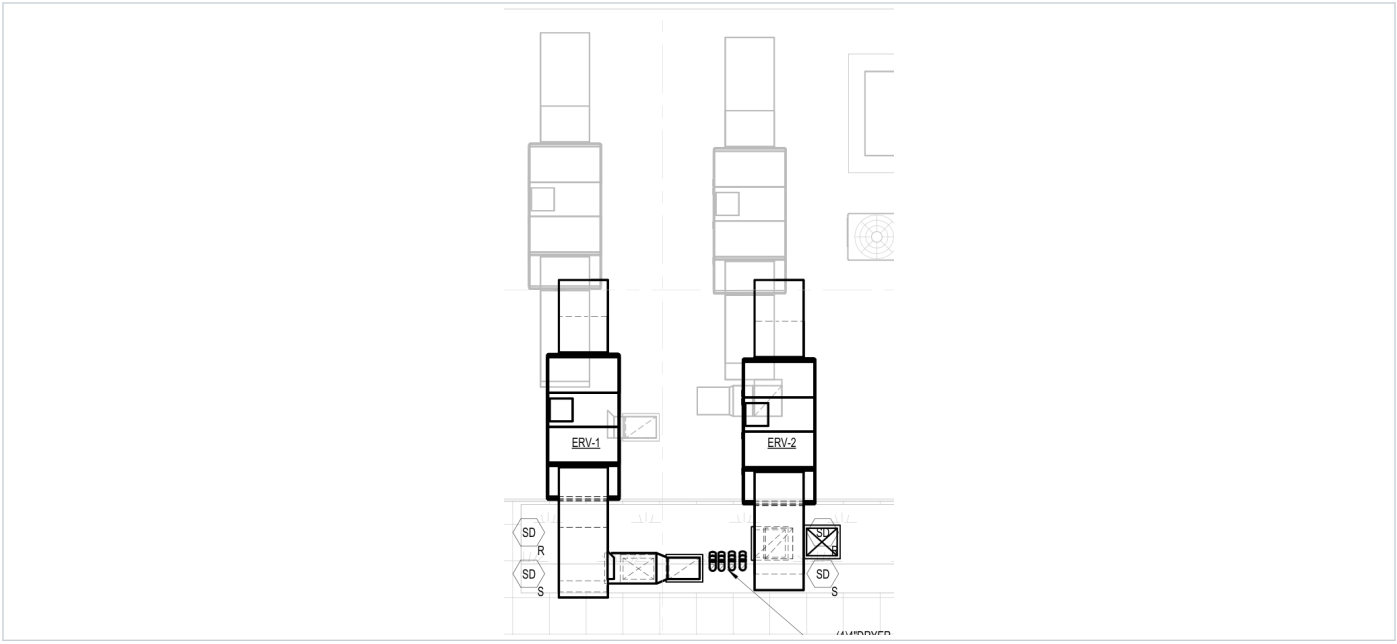
As designed

- Four separate 4" dryer exhaust ducts from the laundry room, terminated 2 ft above the roof

Proposed

- Owner-furnished ventless heat-pump dryers: delete the four dryer exhaust ducts, roof penetrations and terminations

Benefits: No dryer duct cleaning or lint build-up in the risers; fewer roof penetrations; no conditioned air exhausted by the dryers.
Electrical impact: Dryer circuits per the selected dryers (EE)
Approvals / coordination: Owner (laundry equipment); RKTB; condensate drain by plumbing.



M-108.00 Roof - (4) 4" dryer exhaust terminations next to ERV-1 / ERV-2

VE-07 Cellar service rooms: ventilation in lieu of split AC where the load allows

Area: Cellar - AC-COMP and AC-EMR rooms (AC-TEL kept)

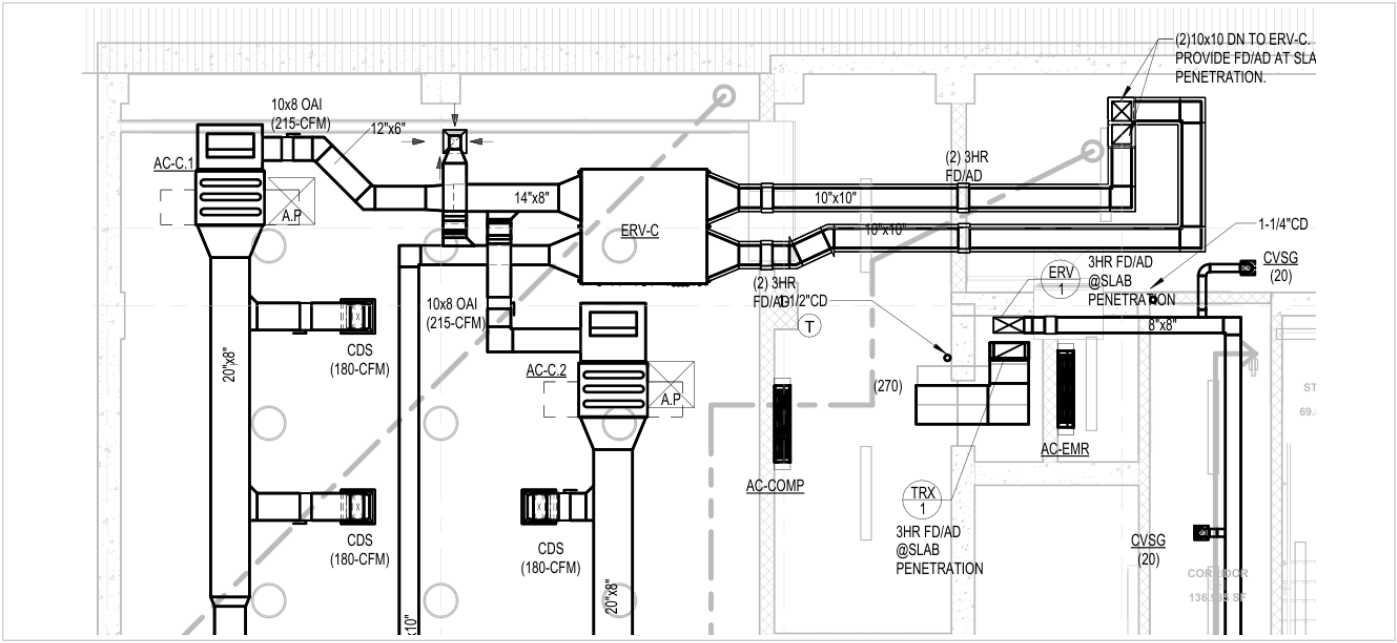
As designed

- Ductless split AC units AC-COMP, AC-EMR and AC-TEL in the cellar service rooms

Proposed

- AC-COMP: if the room is the compactor / trash room, delete the AC; the room is exhausted
- AC-EMR: thermostat-controlled exhaust with transfer / make-up air if the electrical heat load allows
- Telecom room AC-TEL kept as designed

Benefits: Up to 2 fewer split systems with their refrigerant piping and condensate drains.
Electrical impact: Smaller connected load
Approvals / coordination: Altera Engineering (room use and heat loads); EE.



M-100.00 Cellar - ERV-C, AC-C.1 / AC-C.2, AC-COMP and AC-EMR

VE-08 Cellar and 1st floor ventilation: one ERV in lieu of two

Area: ERV-C (cellar) and ERV-CF (1st floor)

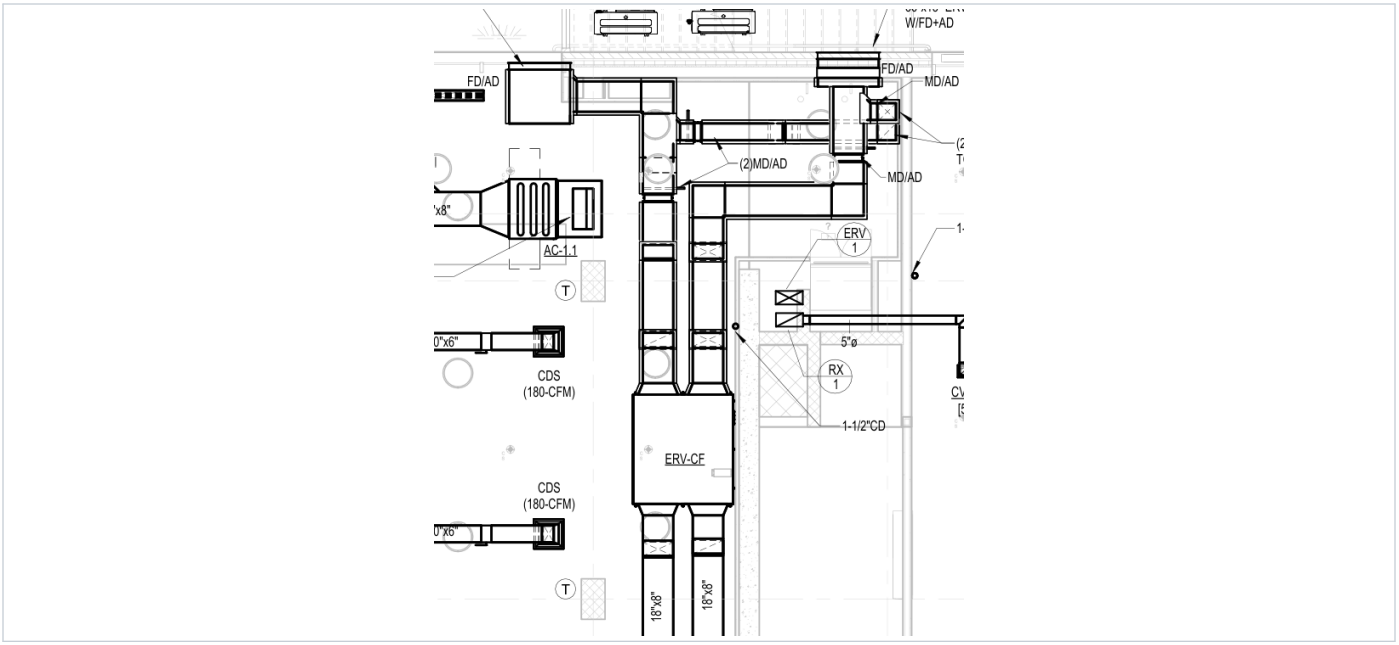
As designed

- ERV-C in the cellar, ducted up through the slab ((2) 10x10 with 3-hour FD/AD) to the 1st floor louvers beside ERV-CF
- ERV-CF on the 1st floor with its own 36"x18" outdoor air and exhaust louvers

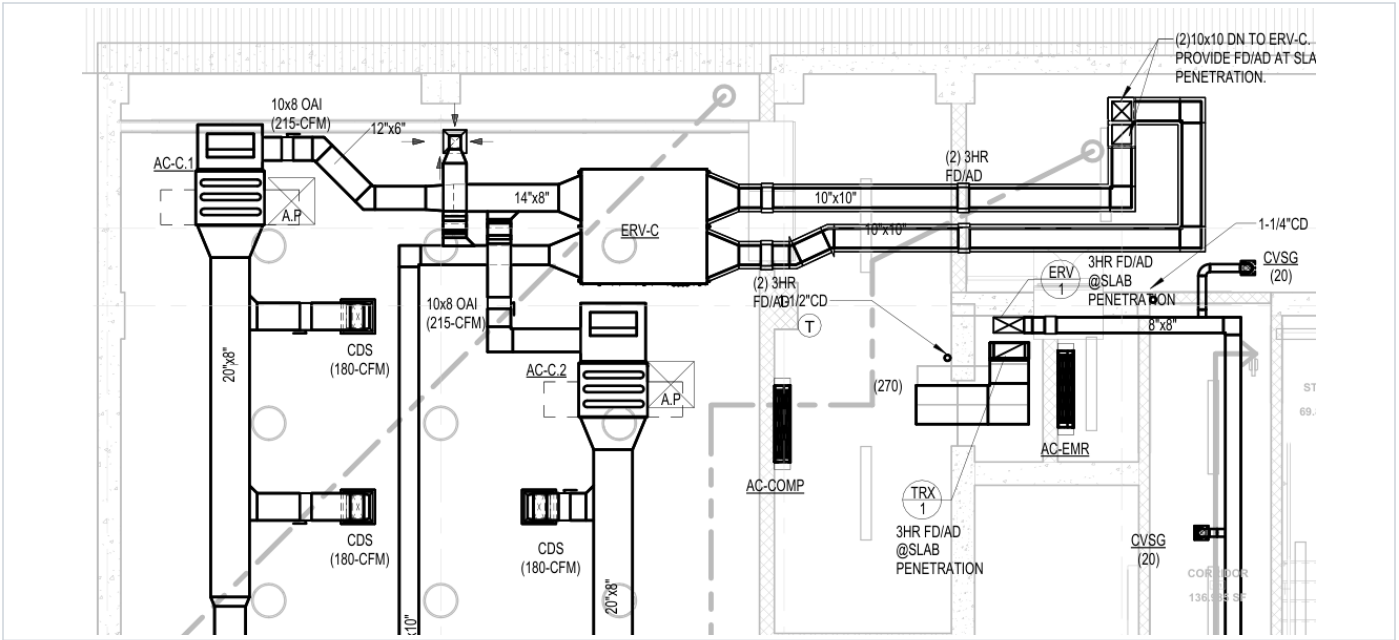
Proposed

- One ERV on the 1st floor sized for both floors, with branch ducts down to the cellar; one set of louvers and dampers. The 3-hour FD/AD at the slab stay

Benefits: One ERV fewer, with its controls, power connection and duct connections.
Electrical impact: One unit fewer
Approvals / coordination: Altera Engineering (airflow, fire separation); owner, if the cellar and 1st floor are separate tenants.



M-101.00 1st floor - ERV-CF and the (2) 10x10 ducts down to ERV-C



M-100.00 Cellar - ERV-C with (2) 3-hour FD/AD at the slab

Coordination items found during the review (RFI to follow)

- The pricing set has floor plans only (M-100 to M-109): no equipment schedules, details, riser diagrams or specification. Quote 3838 used allowance equipment per Procida's direction of 8/28/2026.
- Tag ERV-2 is used for two units (M-102 2nd floor and M-108 roof); tag ACCU-1.2 appears twice on M-101.
- Use of the room served by AC-COMP.
- Fire dampers at the apartment branches into the OA / KX / TX shafts are not shown on the typical floors. For the exhaust shafts we propose steel subducts extending 22" up the shaft with continuous exhaust (NYC Mechanical Code 607.5.5 exception) in lieu of fire dampers; supply shaft per the engineer.
- Is the project Passive House / NYSERDA certified? This affects VE-01 and VE-02.